



Material Handling concepts

March 2025
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Introduction videos

What is material handling? (2,5 min snippet)

https://www.youtube.com/watch?v=PusvVnC_4Uc&t=169s

DHL logistics and inside their warehouses (10 min snippets)

<https://www.youtube.com/watch?v=6EDCnhbUpgE>

Importance of Material Handling

- Material refers to commodities, information, people
- Material handling accounts for:
 - 25% of employees in a typical factory
 - 55% of factory space
 - 87% of production time
- An effective materials handling system can reduce operating costs by 15-30%
- Between 3-5% of material handled becomes damaged
- Causes a lot of industrial accidents
- Objective should be to eliminate & simplify material handling activities
- Important influence on operations design and layout design



Relationship between MH & facility layout

- The relationship between the two design elements involve:
 - Common data required for the design of each activity
 - A common goal to minimize costs
 - Both designs determine the effective use of space
 - Both determine the flow patterns in the facility
- Ideal approach - solve both problems **simultaneously**
- Complexity in practice – requires sequential design with iterations



Definitions of Material Handling

Definition 1

Material handling includes all the basic operations involved in the movement of bulk material, unit loads and individual products in any form, solid, liquid or gas, by means of machinery or by hand, within the boundaries of a place or business

Definition 2

Material handling is the art and science of moving, storing, protecting and controlling material

Definition 3

Materials handling means providing the right amount of the right material at the right place, time, sequence, condition, position and cost by using the right methods



Scope of Material Handling

3 views on scope of material handling activities:

1. Conventional – focus solely on movement of material from point A to B
2. Contemporary – focus on overall movement, develop integrated plan
3. Progressive – total system view, from suppliers to customers

Material handling should:

- Consider the overall flow and develop an integrated plan by considering interrelationships between tasks/activities
- Have a holistic system perspective through integration with manufacturing planning, material management, distribution systems and information management



Objectives of Material Handling

If designed well it can:

- Increase throughput,
- reduce inventories and operating costs

through effective material handling



Auditing current Material Handling processes

Table 5.1 *Material Handling Audit Sheet*

Conditions Indicating Opportunities for Improvement	Condition Observed	Condition Will Require:				
		Supervisor Attention	Management Attention	Analytical Study	Capital Investment	Other Comments
1. Production equipment idle due to material shortage						
2. Material piled directly on floor						
3. In-plant containers not standardized						
4. Operators travel excessively for materials and supplies						
5. Excessive demurrage						
6. Misdirected material						
7. Backtracking of material						
8. Automatic data collection system not used						
9. Excessive trash removal						
10. System not capable of expansion and/or change						
11. No pre-kitting of work						
12. Crushed loads in block stacking						



Material Handling Principles

1. Orientation

- Study the problem to identify existing methods, problems and physical and economical constraints to establish future requirements and goals

2. Planning

- Establish a plan to include basic requirements, desirable options and the consideration of contingencies for all material handling storage facilities

3. Systems

- Integrate economically viable handling and storage facilities into a coordinated system of operations

4. Unit load

- Handle product in as large unit loads as practical/economical



Material Handling Principles (1)

5. Space utilization

- Make effective use of all cubic space

6. Standardization

- Standardize methods and equipment where possible

7. Ergonomic

- Recognize human capabilities and limitations by designing material handling equipment and procedures for an effective interaction with people using the system

8. Energy

- Include energy consumption of material handling systems and material handling procedures when making comparisons or preparing economic justifications



Material Handling Principles (2)

9. Ecology

- Use equipment and procedures that minimize adverse effects on the environment

10. Mechanization

- Mechanize where feasible to increase efficiency and economy

11. Flexibility

- Use methods and equipment that can perform a variety of tasks under a variety of operating conditions

12. Simplification

- Eliminate/reduce/combine movement and equipment

13. Gravity

- Utilize gravity to move material where possible



Material Handling Principles (3)

14. Safety

- Provide safe equipment and methods

15. Computerization

- Consider computerization in handling and storage systems for improved material and information flow

16. System flow

- Integrate data flow with physical flow

17. Layout

- Prepare an operations sequence and equipment layout for all viable system solutions – then select the alternative system which best integrates efficiency and effectiveness



Material Handling Principles (4)

18. Cost

- Compare economic justification in equipment and methods as measured by expenses per unit handled

19. Maintenance

- Prepare a plan for preventative maintenance and scheduled repairs on material handling equipment

20. Obsolescence

- Prepare a long-range and economically sound policy for replacement of obsolete equipment methods with consideration of after-tax life cycle costs

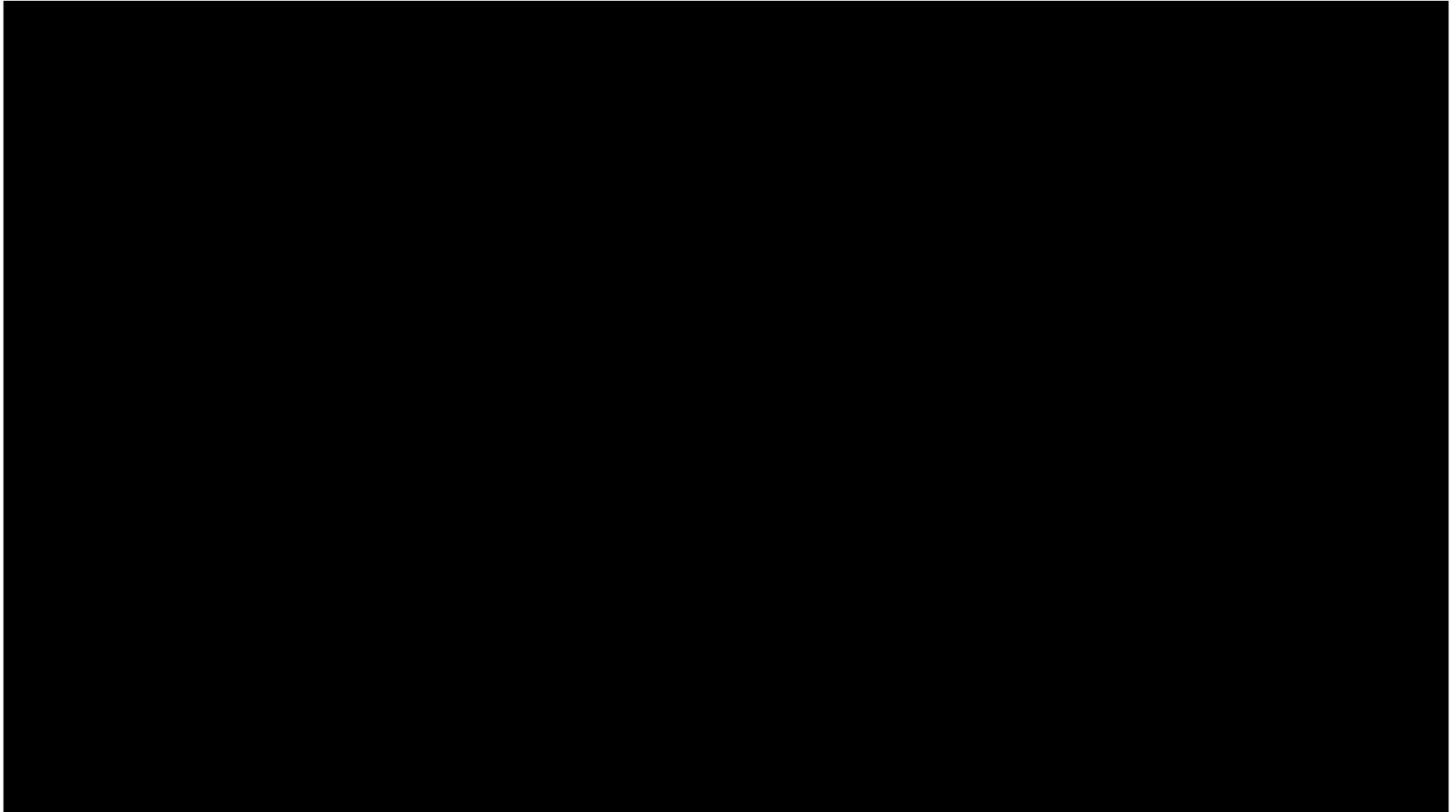


Mechanization of Material Handling

1. Handling by hand - dependent on physical strain
2. Mechanized
3. Mechanization complemented by computers - generate documents and specify movements and processes
4. Automated – minimal human intervention
5. Fully automated – computer supply real-time control, no human intervention (let us have a look at BMW!)



Fully Automated Factory BMW



Cost of Material Handling

- Initial cost of equipment and training
- Operating and Maintenance costs
- Purchasing price for unitizing material of unit loads
- Cost of packaging material
- Cost of material damages

Reduce costs

- Minimize standing time
- Minimize back flow/double handling
- Reduce handling distances
- Plan maintenance procedures
- Correct equipment reduces damages
- Utilize unit loads where possible
- Eliminate unsafe practices
- Standardize
- Use gravity where possible
- Replace obsolete equipment



Engineering Design Process as applied to MH p181

1. Define the objectives and scope of the material handling system
2. Analyze the requirements for moving, storing, protecting and controlling material
3. Generate alternative designs for meeting material handling system requirements
4. Evaluate alternative material handling systems
5. Select the preferred design for moving, sorting, protecting and controlling material
6. Implement the preferred design including the selection of suppliers, training of personnel, installation, debug and start-up equipment and periodic audits of system performance



MHS Design Factors to be considered

- Capital requirements concerning equipment and unitizing of unit loads, availability of funds
- Physical properties of facility and available space
- Management policy in terms of safety and working environment
- **Interdependence** between handling and processing



MHS Design Equation

Material + Movement + Methods = Recommended Systems

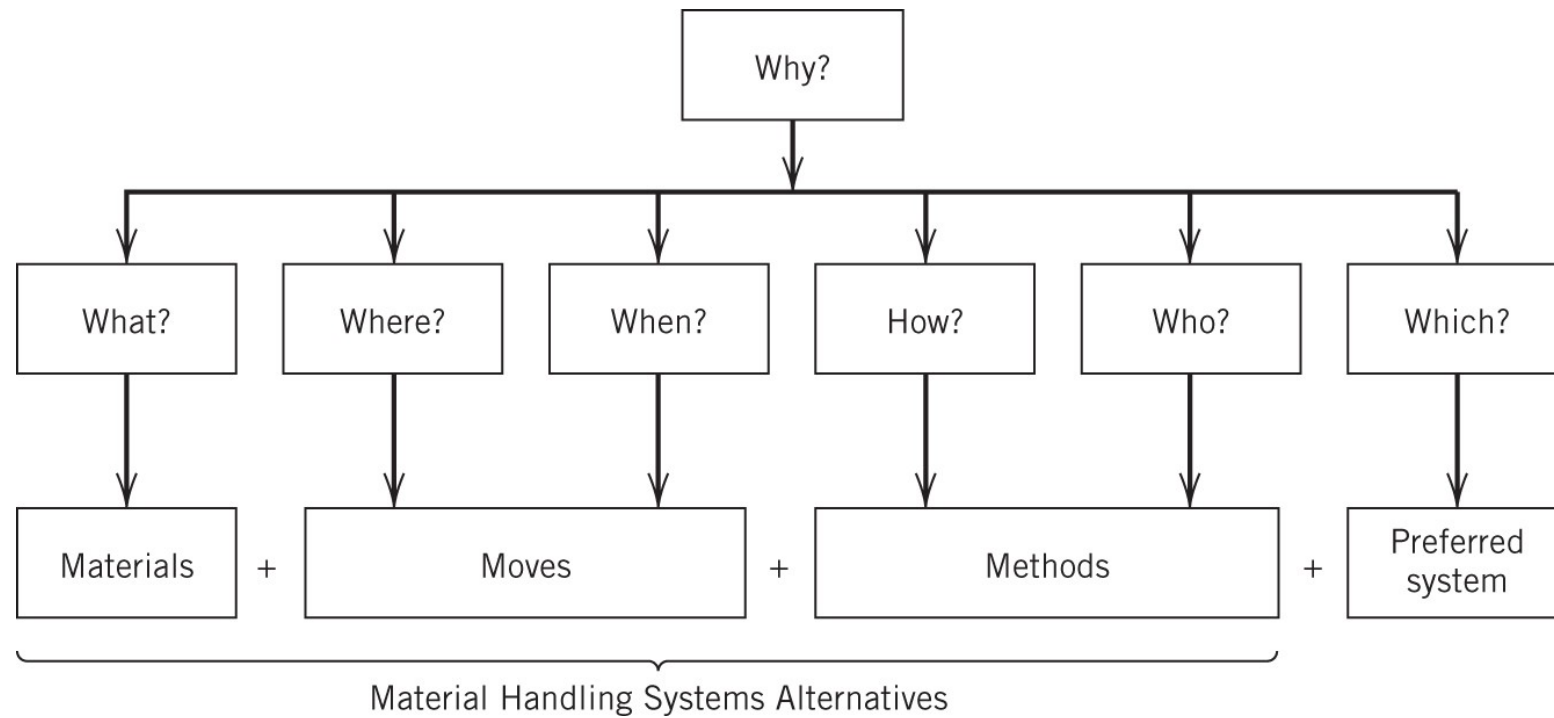


Figure 5.1



Problems In the Design of a MHS

- Interdependency between materials handling and layout
- **Wide variety of available equipment!!!! See Appendix 5B for examples**
- Some characteristics of equipment are difficult to quantify (for example flexibility, mobility and effort)
- Objectives other than minimum cost (utilization, safety, maintenance etc.)
- Application of an analytical model usually requires complex calculations with no guarantee for an optimal system design
- Some information required for the design of the system (e.g. movement time and equipment running cost) cannot be determined accurately, except when already in use



Characteristics of an efficient MHS

- Combines handling with processing where possible
- Automation where possible
- Safe
- Provides protection for material
- Standardization in types of equipment
- Maximum utilization of equipment
- Minimum backtracking, cross flows and handling
- Minimum congestion or delays
- Economical



Unit Load Principles

- Move or handle items or material grouped *together* by some means
- Capability to handle more items at a time thus reducing the number of trips, handling costs, loading and unloading times and product damage:
- Types of containers and unitizing equipment include:
 - Containers – Pallets, skids and skid boxes, tote pans
 - Unitizers – Straps/stretch-wrap, palletizers



Factors Determining the Size of Unit Load

- Type, mass, size and shape of materials/items to be handled
- Quantity and frequency handled
- Susceptibility to damage
- Extent of handling (distance etc.)
- Environment (available space, aisle etc.)
- Reconcilability with the total logistics system
 - Other unit loads in system
 - Material handling and transport equipment in system
 - Method of receiving, storing, shipping and handling
- Cost of unitizing load
- Additional processes in unitizing a unit load such as stacking and protection materials



Material Handling Equipment

The type of equipment is influenced by the properties of material, volume, physical environment and nature of processes.

Classification according to reach

- Unlimited – industrial vehicles
- Area limited – overhead crane
- Line limited – conveyers
- Position limited – stacker crane

Classification according to type of equipment

- Conveyers – frequent movement
- Monorails, cranes and hoists – interrupted transferring
- Industrial vehicles – internal transport

There is a variety of equipment – standard vs purpose-made



Factors Determining Equipment Choice

Material

- Type, size, volume, shape, weight, unit load of materials

Movement requirements

- Distance, frequency, path, aisle width, loading/unloading mechanism

Storage requirements

- Area, height, shape and obstructions in storage facilities, storage equipment and policies

Cost

- Capital and operating cost of equipment, interest rates, depreciation and lifespan

Other factors

- Flexibility and obsolescence



See both TEXTBOOKS for variety of equipment ideas

